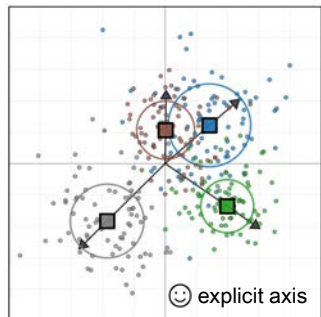


Subspace (Representation)



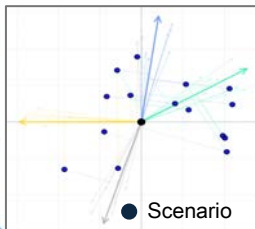
Embedding PCA space



Domain Subset Selection

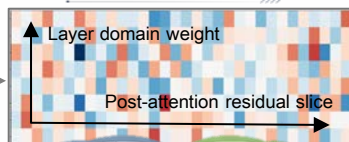
(1) Orthogonality (2) Coverage

Domain-wise Weight

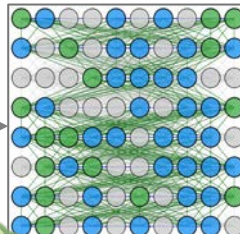


Mechanism: Representation-Parameter Coupling

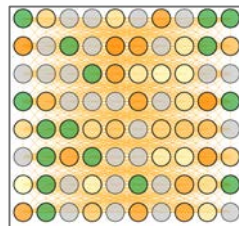
Probe



Head-wise Pruning



Pathway (Parameter)



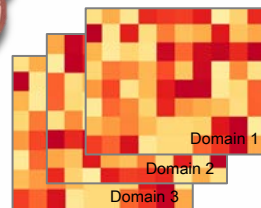
③ PSP-online

Kept head
Pruned head
Whitelist (kept)

Head-wise Score corresponding to specific scenario

② PSP-offline

Cached Head Importance



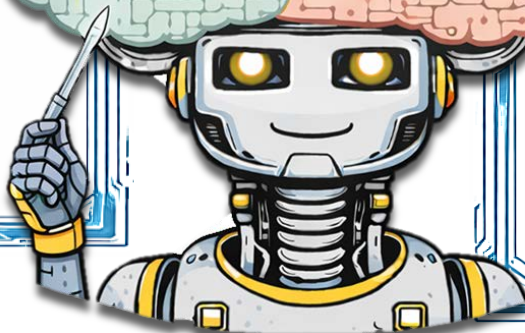
Head-wise Importance corresponding to Domain-K

Cached Whitelist

General-purpose capacity head
Semantically highly relevant head



LLM Dense Backbone



Subspace-Pathway Coupling Block conceptually

Pathway highly related to current scenario